

RESEARCH ARTICLE

Human–wildlife coexistence needs more evidence-based interventions to reduce the losses of crops, livestock and fishery catches

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Abstract

1. Evidence-based interventions designed to reduce wildlife-caused losses are essential for human–wildlife coexistence. The lack of systematic summarization of research effort and evidence makes it challenging for researchers, managers and policymakers to prioritize interventions for evaluation and implementation.
2. Here, we compiled experimental case studies of nonlethal technical interventions designed to reduce the losses of crops, livestock and fishery catches caused by terrestrial carnivores, elephants, farmland birds and marine fauna worldwide. Then, we summarized the research effort and the performance of interventions by their sensory stimuli and target animals.
3. We found that: (i) 54 of 88 interventions included in this study had statistically effective evidence, where only 39% (21/54) were evaluated with more than three experiments; (ii) physical-, sound-, chemical- and light (or visual) -based interventions were the most in numbers and their performance varied greatly; (iii) farmland birds, seabirds and cetaceans were the most studied animal groups while there are only a few experiments for elephants; and (iv) the interventions for marine fauna generally had no impact on the target catch of fisheries.
4. *Syntheses and applications:* Our results indicated that collective effort is needed to further evaluate interventions using various sensory stimuli and launch incentive programs to motivate the implementation of interventions, particularly related to marine fauna conservation. Our synthesis could be helpful for stakeholders

to tackle the negative human-wildlife interactions outlined as Target 4 of the Kunming–Montreal Global Biodiversity Framework.

KEYWORDS

crop raid, fishery loss, human–wildlife conflict, livestock depredation, mitigation

1 | INTRODUCTION

Humans and wildlife coexist on a substantial scale on the earth. One of the most challenging issues for coexistence is human–wildlife conflict (Nyhus, 2016). In practice, the ‘conflict’ mainly refers to wildlife-caused harm to humans and the induced harm to wildlife (e.g. persecutions; Fisher, 2016; Peterson et al., 2010). These ‘conflicts’ are widespread and affect many threatened species and those with high socio-ecological values (Huang et al., 2023; Montgomery et al., 2022; Nyhus, 2016; Tixier et al., 2020). These animals can cause substantial costs to humans including human casualties, and losses of crops, livestock and fishery catches. Therefore, mitigating such ‘conflicts’ could be key to achieving many Sustainable Development Goals (e.g. no poverty and life on land) and Target 4 of the Kunming–Montreal Global Biodiversity Framework (<https://www.cbd.int/gbf/targets/4>).

Various interventions have been developed and deployed to mitigate property losses (Huang et al., 2023; Montgomery et al., 2022; van Eeden, Crowther, et al., 2018). According to the target objectives, these interventions could be broadly classified as wildlife-oriented (e.g. lethal control and repellents), habitat-oriented (e.g. alternative crops or land use modification) and human-oriented (e.g. space or time enclosure for human activities; Nyhus, 2016). Unless quantitative evaluations are available, the effects of interventions often cannot be precisely perceived by implementers and potential investigators, especially in the case of crops and fishery catches.

Many case studies have been conducted to provide quantitative evidence of interventions (Huang et al., 2023; Montgomery et al., 2022; van Eeden, Crowther, et al., 2018). For designing a solid case study, there are four key elements to be considered, that is the problem (i.e. the animal and properties involved), the intervention, the setting up of experimental comparisons and the outcome measurement (Nishikawa-Pacher, 2022). Optimally, the setting up, such as the selection and assignment of experimental units (e.g. crop fields, herds, and fishing nets) to treatment or control groups, should be controlled to reduce various confounding biases in the wild for causal inference (Khorozyan, 2022). Meanwhile, the outcome of interventions should be measured directly on the affected properties to inform further implementations. However, currently, many case studies were observational without any control of the setting up and the outcome was indicated by indirect measures (e.g. contact and deter rate; Khorozyan, 2022; Montgomery et al., 2022). These simplifications weaken the inference power of case studies for implementations (Christie et al., 2019; Khorozyan, 2022; Treves et al., 2016).

Furthermore, it was widely recognized that the effects of interventions can vary across conditions (e.g. species involved, resource distribution pattern and management practices; Huang et al., 2023; Montgomery et al., 2022; van Eeden, Eklund, et al., 2018). Therefore, synthesizing the evidence from several sites and experiments can provide useful insights for potential scale-up. To date, six syntheses (Table S1) have systematically compiled lists of case studies to evaluate the effects of interventions for the four most representative conflicts involving large terrestrial carnivores, elephants, farmland birds and marine megafauna (Huang et al., 2023; Huang, Rice, et al., 2024; Khorozyan, 2022; Lorand et al., 2022; Montgomery et al., 2022; van Eeden, Eklund, et al., 2018). These syntheses included case studies greatly varying in the study designs (e.g. observational and experimental studies) and the outcome measurements (Table S1). Furthermore, these syntheses used different metrics to measure the effects of interventions, including the quantitative (e.g. Hedge's *d* and *g*) and the qualitative (e.g. effective, ineffective and counterproductive; Borenstein et al., 2021). These syntheses also used different summarization techniques (e.g. meta-analysis and counting) to summarize these effects for an intervention or a category of interventions (Borenstein et al., 2021). Consequently, the research effort and the amount of effective or counterproductive evidence for interventions are incomparable across interventions and animal groups, which hinders researchers, managers and policymakers from prioritizing interventions for evaluations and implementations.

On the other hand, the effects of interventions may depend on the target sensory modality of animals. A few senses are unique for some animals, such as the echolocation for cetaceans and the electro-sense for elasmobranchs (Lucas & Berggren, 2022). As such, the types of interventions may be essentially distinct among animal groups. For example, using rare earth metal on the hook is tailored to deter elasmobranchs and attaching bead chains to fishing nets is assumed to increase the detectability of fishing nets by cetaceans (Lucas & Berggren, 2022). Meanwhile, many animals (e.g. terrestrial carnivores, elephants, birds and marine fauna) can broadly perceive the stimuli of light, sound and chemicals (Elmer et al., 2021; Garvey et al., 2020; Lucas & Berggren, 2022). Therefore, some sensory stimuli can be applied to interventions for multiple animal groups, such as the toriline to deter seabirds and reflecting ribbon for farmland birds, flashing lights to deter terrestrial carnivores in ranches and illuminators to reduce the fishery–marine fauna interactions and loud sound to deter farmland birds and cetaceans, though the devices used in these interventions are ostensibly different. In this context, understanding the sensory stimuli of interventions may help to identify the directions to design new interventions (Blackwell et al., 2016; Elmer et al., 2021; Garvey et al., 2020; Lucas

& Berggren, 2022). However, previous relevant studies are often narrative reviews based on non-systematic compilations of case studies (Barrett et al., 2019; Blackwell et al., 2016).

In this study, we attempted to summarize the experimental effort for nonlethal technical interventions designed to reduce the losses of crops, livestock and fishery catches caused by terrestrial carnivores, elephants, farmland birds and marine fauna (i.e. pinniped, cetaceans, seabirds and elasmobranchs). Here, an 'experiment' means that an intervention is deliberately introduced to observe its causal effects and the setting up of experimental comparisons is somehow controlled (Cook et al., 2002; Dunn, 2021). The outcome measurements are on the affected crops, livestock and fishery catches. Then, we summarized the amount of statistically effective and counter-productive evidence by interventions, their sensory stimuli and target animals. Our results may be instrumental for stakeholders to tackle the negative interactions outlined in Target 4 of the Kunming-Montreal Global Biodiversity Framework and highlight the interventions for further evaluations and implementations.

2 | MATERIALS AND METHODS

2.1 | Compilation of the list of case studies

We collected case studies from the six syntheses (Huang et al., 2023; Huang, Rice, et al., 2024; Khorozyan, 2022; Lorand et al., 2022; Montgomery et al., 2022; van Eeden, Eklund, et al., 2018), which were based on systematic search schemes and provided the most comprehensive lists of case studies to our knowledge. For terrestrial carnivores, van Eeden, Eklund, et al. (2018) compiled 114 case studies based on four systematic reviews (Eklund et al., 2017; Miller et al., 2016; Treves et al., 2016; van Eeden, Crowther, et al., 2018); Lorand et al. (2022) compiled 103 studies; and Khorozyan (2022) compiled 244 studies to summarize the study designs. For elephants, Montgomery et al. (2022) compiled 95 studies. For farmland birds, Huang et al. (2023) compiled 48 studies. For marine fauna, Huang, Rice, et al. (2024) compiled 119 studies. After removing the duplicated studies, 594 case studies were left for the initial list (Figure S1).

2.2 | Inclusion criteria

These six syntheses had different inclusion criteria about target animal groups, types of interventions, study designs, reported data and statistical approaches (Table S1). According to our research objectives, we set six criteria to make the research effort comparable. If a case study did not meet one of the following criteria, we would exclude it from further review.

- (i) *Language*. The studies had to be published in the working languages of the reviewing team (i.e. English and Chinese).
- (ii) *Problem*. The involved animals should be wild and free-ranging animals. The affected properties should be crops, livestock and

fishery catches. Notably, while terrestrial animals are often intentionally controlled, marine megafauna usually is entangled or hooked unintentionally when they feed on baits and captured fishes, leading to injuries or drowning and further damage to fishing gear (Gilman et al., 2023; Wilson et al., 2014).

- (iii) *Nonlethal technical intervention*. Interventions must be nonlethal technical measures and adjusting fishing or farming time were excluded.
- (iv) *Comparisons*. While the random setup of a strict experiment is often impossible for case studies in this research field, we required the setup of experimental comparison to be somehow controlled to reduce various confounding biases in the wild.
- (v) *Outcome*. The effects of interventions on properties should be measured. The measurements include weight of crop yield, area or weight of crop damage, number of livestock lost, and weight or number of captured fish. We excluded experiments based on indirect behavioural indicators, such as the number of visitations, contact rate (e.g., contact rate between seabirds and gears near baited hooks), and respondents' perceived effectiveness. For translocations, the change of losses of the pre-release site and the post-release site should be measured.
- (vi) *Statistical test*. Significance testing of the outcome between the treatments and controls should be reported or is possible to calculate from the reported data.

In general, to include more types of interventions, we relaxed the inclusion criteria on case studies about the reported data compared to previous meta-analysis (Huang et al., 2023; van Eeden, Crowther, et al., 2018). The meta-analysis approaches are often used to evaluate the overall impact of an intervention based on quantitative effect size (e.g. Hedge's *d* and *g*) calculated from detailed data (e.g. the mean, standard deviation and sample size) obtained from multiple case studies and may detect statistically significant effects of interventions, even when synthesizing insignificant case studies of low statistical power (Borenstein et al., 2021; Koricheva & Gurevitch, 2013). However, case studies that do not report the required data for calculating the effect size were usually excluded, such as the studies for fyke net modification for pinniped (Königson et al., 2007) and umbrella-and-stones system for seabirds (Brown et al., 2010). Besides, many interventions were evaluated only by a single experiment (or one effect size), such as using decoy fields (Hagy et al., 2008), seed coat (Decker et al., 1990) and caffeine (Avery et al., 2005) for farmland birds, were often excluded from the meta-analysis (Huang et al., 2023). Meanwhile, we tightened the inclusion criteria to that of previous narrative reviews on the study designs and outcome measurement.

With our different criteria from previous syntheses, we revisited the inclusion process of Huang et al. (2023) and Huang, Rice, et al. (2024) and identified an additional 13 and 8 studies, respectively, which did not meet the previous inclusion criteria of the meta-analysis but met the criteria of this study. Therefore, there were 615 case studies in the final list and 605 of them are available online. The inclusion process of this study was documented in the Supplementary Material (Figure S1).

2.3 | Data extraction and compilation

For each case study, we extracted information on the types of interventions and their outcomes. An intervention was recognized mainly based on the combination of the target animal groups and device or gear its used. We classified intervention outcomes as either statistically effective, insignificant or counterproductive according to the original case studies. Statistically effective means that an intervention showed statistical significance in reducing the loss or increasing the yield of properties; counterproductive means that an intervention showed statistical significance in increasing the loss or reducing the yield of properties; and statistically insignificant means that an intervention had no statistically significant impact on the properties but this does not necessarily mean the effect of an intervention is absent due to statistical powers (Borenstein et al., 2021). For studies that reported the effects of interventions on multiple livestock (e.g. sheep, goat and cattle) or target fishes, we extracted the overall statistical test, if available. We treated the level of 0.05 as statistically significant.

We compiled the data about the environment (i.e. the terrestrial or marine), animal groups, sensory stimuli and geographical information (e.g. countries and major fishing areas). According to management and research practices, we classified the animals into terrestrial carnivores, elephants, farmland birds, seabirds, cetaceans, pinnipeds and elasmobranchs. The sensory stimuli could be light (or visual, e.g. flashing light and movable predator models), sound (e.g. acoustic pingers), chemical (e.g. anthraquinone), electricity (e.g. electric wire), magnetic (e.g. permanent magnet), multi-sensory stimuli (multimodality, hereafter, e.g. alternative crops or baits) and physical approach (e.g. fortified fences and line weightings to avoid depredations) and combinations of the former stimuli (e.g. toriline as the light-physical and guardian animals as the multimodality-physical; Figure 1 and Table S2). We assigned each study location (if available) to the countries' territory, Economic Exclusive Zone of a specific country, and the major fishing area as defined by the Food and Agriculture Organization.

We classified feeding stations or decoy fields as physical approaches that lead wildlife away. We classified the bird perch and aerial-applied herbicide to habitat as two indirect interventions because they relied on the presence of aggressive birds and the removal of weeds at roost habitats. We only included statistically effective interventions when a study aims to examine the outcome of multiple similar interventions without making any assumptions about which one would work. Examples of such interventions included the colours of fruit-protective bags (Mutisya et al., 2016; Wen et al., 2016).

2.4 | Data analysis

We used Pearson's chi-squared test to check whether the number of experiments of interventions by sensory stimuli, animal groups and environments were equally distributed in a sample group.

3 | RESULTS

Experiments designed to evaluate the interventions were geographically biased. From 147 included case studies, we extracted information from 244 experiments, of which 54% and 46% for terrestrial and marine fauna, respectively. For terrestrial fauna, about half of the experiments were conducted in a single country (i.e. the USA, $n=65$). For marine fauna, most experiments were conducted in the Northeast Atlantic ($n=36$), Northeast Pacific ($n=18$), Mediterranean and Black Sea ($n=17$), and Western Central Pacific ($n=10$). Meanwhile, most of the experiments in the Economic Exclusive Zones were conducted in the Western world (e.g. Western European countries, the USA, and Australia, Figure 2).

3.1 | Research efforts for interventions

The amount of evidence for an intervention was generally scarce. In total, 88 interventions were evaluated (Figure 1). The average number of experiments was three for the 88 interventions (range from 1 to 22), of which 27% (24 out of 88) were evaluated in three or more experiments (nodes with green linking lines in Figure 1) while the 56% (49 out of 88) were evaluated only by a single experiment.

The physical approaches were the mostly applied in the included interventions. The number of experiments was unequally distributed ($\chi^2=316.95$, $df=11$, $p<0.01$) across sensory stimuli. The physical, sound, chemical, light and light-physical were evaluated in 83, 40, 36, 27 and 24 experiments, respectively. In contrast, there were very few experiments for the electricity-physical ($n=3$; electric fences for carnivores), electricity ($n=2$; electric wire for pinniped and rare earth metal for elasmobranchs), magnet ($n=2$; permanent magnet for elasmobranchs) and chemical-physical ($n=1$; chilli pepper fence for elephants; Figure 3 and Table S3).

There were very few experiments for elephants. The number of experiments was unequally distributed ($\chi^2=159.59$, $df=6$, $p<0.01$) across animal groups, among which farmland birds ($n=101$), seabirds ($n=35$) and cetaceans ($n=30$) were the most studied. In contrast, the interventions for elephants were only tested in seven experiments.

3.2 | Outcome of interventions

Many interventions had statistically effective evidence for the contexts where they were tested, but their performances often varied across contexts. In the data set, 61% of interventions ($n=54$) had statistically effective evidence while only five interventions with three or more experiments showed consistent performance and the 10% ($n=9$) had counterproductive evidence, which were all observed for the marine fauna.

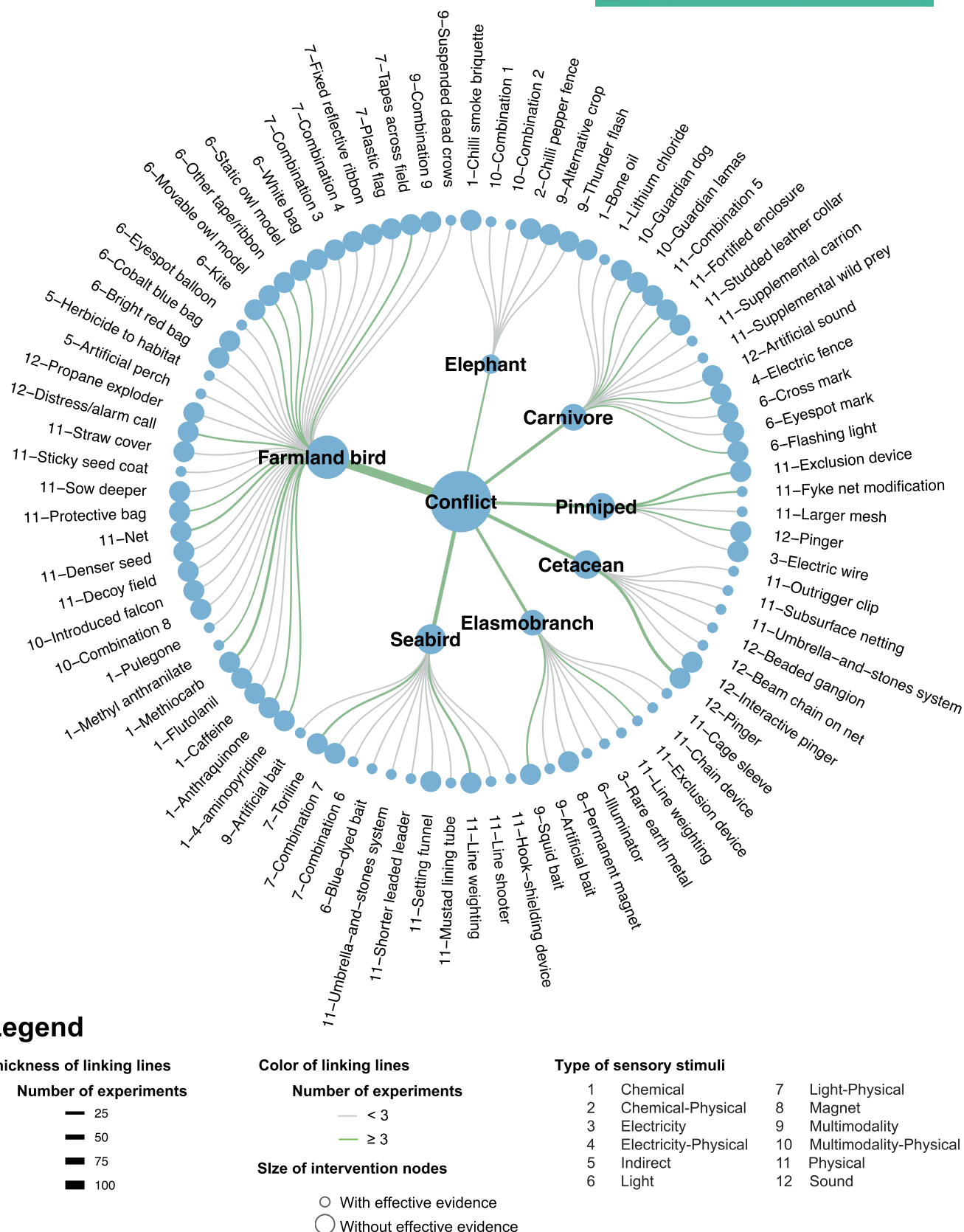


FIGURE 1 The topography of interventions to reduce wildlife-caused property losses. The bigger nodes of animal groups and the thicker linking lines indicate a greater number of experiments; the green-coloured linking lines mean that the number of experiments is greater than or equal to three; and the bigger nodes of intervention indicate that the intervention had statistically effective evidence. The prefix number in the label of intervention represents the different types of sensory stimuli. The combination of multiple numbers (e.g., 1–9) means that the interventions are a combination of multiple interventions (Table S2).

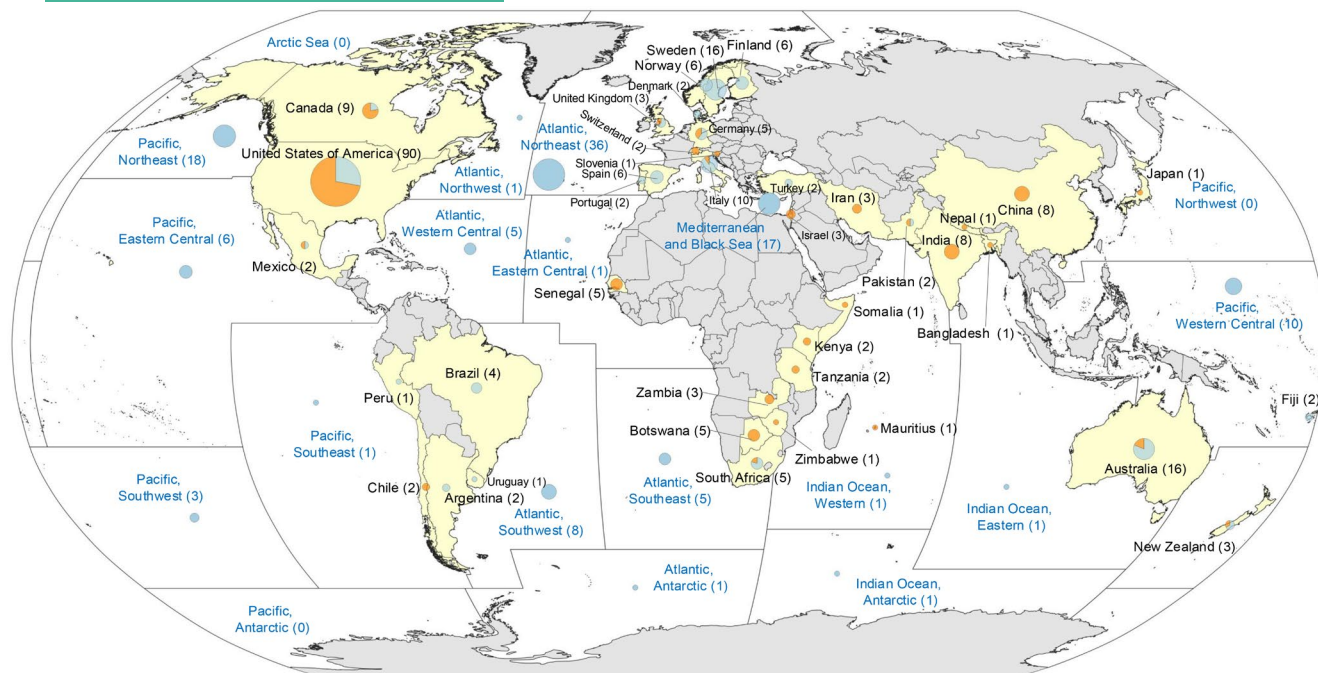


FIGURE 2 Number of experiments of interventions (the value in parentheses) by countries' territory, Exclusive Economic Zones of a specific country, and the major marine area as defined by the Food and Agriculture Organization. The size of the pie chart is proportional to the number of experiments for terrestrial (orange) and marine (blue) fauna.

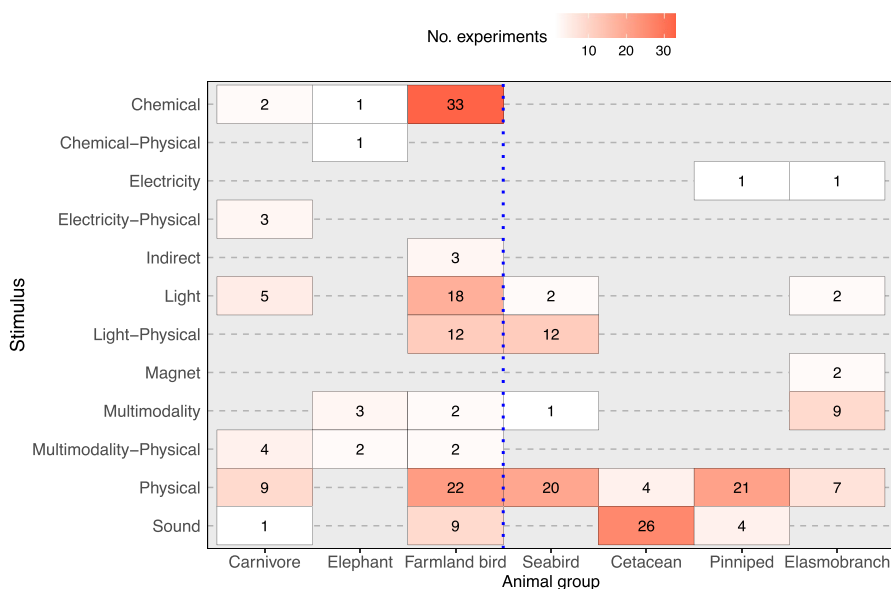


FIGURE 3 The number of experiments by stimuli and animal groups. The data on the left side of the vertical blue dash line is for terrestrial fauna and the right side is for marine fauna.

3.2.1 | Farmland birds

Currently, some interventions, including the use of protective bags on fruits, tapes and ribbons, distress/alarm calls, and chemicals causing physiological disorders (i.e. 4-aminopyridine, methiocarb and anthraquinone) showed a relatively higher proportion of experiments reporting statistically effective evidence to reduce the losses (Table 1). Meanwhile, there was no statistically effective evidence to support some interventions, such as the use of

methyl anthranilate, eyespot balloons, suspended dead birds and artificial perches.

3.2.2 | Carnivores and elephants

Physical barriers, including electric fences, fortified enclosure and reactive night-kraaling informed by early-warning systems, were often statistically effective in reducing the livestock depredation caused by

TABLE 1 Experimental evidence of interventions for crop losses caused by farmland birds.

Stimuli	Intervention	No. experiments	% stat. –	Country	Reference
Physical		22	59%		
	Decoy field	1	100%	USA	Hagy et al. (2008)
	Denser seed	2	100%	Israel	Benjamini (1980)
	Protective bag	4	100%	China	Tran et al. (2015)
	Sow deeper	1	100%	India	Firake et al. (2016)
	Straw cover	2	100%	India	Firake et al. (2016)
	Net	11	27%	Canada, India, Mauritius, Senegal, USA	Bruggers and Ruelle (1982), Conover (1982), Pagay et al. (2013), Firake et al. (2016) and Oleksy et al. (2021)
	Sticky seed coat	1	0%	USA	Decker et al. (1990)
		18	83%		
	Bright red bag	1	100%	China	Wen et al. (2016)
Light	Cobalt blue bag	1	100%	China	Wen et al. (2016)
	Kite	5	100%	USA, UK	Conover (1982), Hothem and DeHaven (1982), Conover (1984), Haque and Broom (1985) and Seamans et al. (2002)
	Movable owl model	3	100%	USA	Conover (1985)
	Other tape/ribbon	3	100%	Iran, Pakistan	Mobini and Khaleghizadeh (2007) and Akram et al. (2013)
	White bag	1	100%	China	Wen et al. (2016)
	Static owl model	3	33%	USA	Conover (1985)
	Eyespot balloon	1	0%	USA	Stickley Jr et al. (1972))
		9	78%		
	Propane exploder	2	100%	USA	Stickley Jr et al. (1972) and Conover (1984)
	Distress/alarm call	7	71%	Switzerland, USA	Naef-Daenzer (1983), Houk et al. (2004), Delwiche et al. (2007), Coates et al. (2010) and Khan et al. (2013)
Sound		33	67%		
	Caffeine	1	100%	USA	Avery et al. (2005)
	Flutolanil	1	100%	USA	Werner et al. (2008)
	4-aminopyridine	6	83%	USA	Stickley Jr et al. (1972), Besser and Guarino (1976), Dolbeer et al. (1976), Besser et al. (1984); Conover, 1984 and Knittle et al. (1988)
	Methiocarb	15	73%	Australia, Israel, USA	Stickley Jr and Guarino (1972), Dolbeer et al. (1973), Ingram et al. (1973), Stickley Jr (1973), Stone et al. (1974), Mott et al. (1976), Bailey and Smith (1979), Benjamini, 1980, Conover (1982), Holler et al. (1982), Tobin and Dolbeer (1987), Tobin et al. (1989) and Tobin et al. (1991)
	Anthraquinone	6	67%	USA	Avery et al. (1998), Cummings, Avery, et al. (2002), Cummings, Pochop, et al. (2002), Kandel et al. (2009), Curtis et al. (2019) and Werner et al. (2019)
	Methyl anthranilate	3	0%	Germany, USA	Avery et al. (1996), Werner et al. (2005) and Esther et al. (2013)
	Pulegone	1	0%	Germany	Esther et al. (2013)

(Continues)

TABLE 1 (Continued)

Stimuli	Intervention	No. experiments	% stat. –	Country	Reference
Multimodality		2	50%		
	Combination 9	1	100%	USA	Cummings et al. (1986)
	Suspended dead crows	1	0%	Switzerland	Naef-Daenzer (1983)
Light-physical		12	67%		
	Combination 3	1	100%	India	Firake et al. (2016)
	Combination 4	1	100%	India	Firake et al. (2016)
	Fixed reflective ribbon	1	100%	India	Firake et al. (2016)
	Tapes across field	7	57%	USA	Dolbeer et al. (1986), Tobin et al. (1988) and Tipton et al. (1989)
	Plastic flag	2	50%	Japan, USA	Mason et al. (1993) and Lane and Nakamura (1996)
Multimodality-physical		2	100%		
	Combination 8	1	100%	USA	Radtke and Dieter (2011)
	Introduced falcon	1	100%	New Zealand	Kross et al. (2012)
Indirect		3	0%		
	Artificial perch	2	0%	Australia, Bangladesh	Ali et al. (2017) and Peisley et al. (2017)
	Herbicide to habitat	1	0%	USA	Linz et al. (1995)

Note: There was no counterproductive case for these interventions. The '% stat. –' represents the proportion of experiments reporting statistically effective evidence to reduce the losses.

carnivores (Table 2). For elephants, there was no statistically effective evidence for two community-based interventions (Table 2). Promisingly, some simple interventions showed potential to reduce the damage, including the use of protective leather collars, painted marks on cattle, flashlights with or without sound, and chilli-based approaches (Table 2).

3.2.3 | Marine fauna

Most interventions for marine fauna generally had no impact on the target catch of fisheries, though the use of setting funnel, toriline, squid bait and acoustic pinger showed potential to reduce the losses caused by seabirds, elasmobranch and pinnipeds (Table 3). The acoustic pingers, with the highest number of experiments, were usually statistically insignificant in reducing the losses. Meanwhile, some interventions may cause counterproductive effects to fisheries, including the use of beam chains on nets, subsurface netting and umbrella-and-stones systems for cetaceans, exclusion devices and rare earth metal for elasmobranchs, exclusion devices and fyke net modification for pinnipeds, artificial baits and umbrella-and-stones system for seabirds (Table 3). Notably, using exclusion devices for pinnipeds had both statistically effective and counterproductive evidence.

4 | DISCUSSION

Human-wildlife 'conflict' is a global issue with detrimental effects on people and wildlife (Nyhus, 2016). Based on the available

experimental evidence worldwide, we summarized the research efforts and the outcome of nonlethal technical interventions designed to reduce the losses of crops, livestock, and fishery catches caused by terrestrial carnivores, elephants, farmland birds, and marine fauna, respectively. We identified two major challenges for the widespread implementations of interventions: (i) the amount of evidence for interventions was generally scarce and (ii) most interventions for the marine fauna were not found to reduce fishery losses. The synthesized evidence would be valuable for researchers, wildlife managers and conservation policymakers to prioritize interventions for evaluations and implementations.

More experimental effort is needed for many promising but insufficiently evaluated interventions. In the wild, various biological (e.g. life stage, sex, natural food availability), environmental (e.g. climate and terrain) and anthropogenic (e.g. gear maintenance, agricultural practices and damage monitoring) factors affect the measured outcome of interventions (Blackwell et al., 2016; Delord et al., 2009; Huang et al., 2018; Nyhus, 2016; Songhurst, 2017). As such, the outcomes of an intervention can vary among sites and multiple experiments are essentially needed to evaluate the consistency of the outcome of interventions and form the best practices for potential scaling-up. In our data set, among the 54 interventions with statistically effective evidence, only 39% of them were evaluated with three or more experiments. Thus, many interventions would require more experiments for further evaluations, including studded leather collars (Khorozyan et al., 2020) and artificial eyespots on livestock (Radford et al., 2020) for terrestrial carnivores, alternative crops and chilli-based interventions for elephants (Gross et al., 2016; Karidozo

TABLE 2 Experimental evidence of interventions for livestock depredation by carnivores and crop losses by elephants.

Taxa	Stimuli	Intervention	No. experiments	% stat. –	Country	Reference
Carnivore	Physical	Combination 5	9	56%	Botswana	Weise et al. (2019)
		Studded leather collar	1	100%	Iran	Khorozyan et al. (2020)
		Fortified enclosure	5	60%	Botswana, China, Tanzania	Kissui et al. (2019), Lichtenfeld et al. (2015), Papworth et al. (2014) and Weise et al. (2018)
		Supplemental carrion	1	0%	Slovenia	Kavčič et al. (2013)
		Supplemental wild prey	1	0%	Canada	Morehouse and Boyce (2017)
	Light	Cross mark	1	100%	Botswana	Radford et al. (2020)
		Eyespot mark	1	100%	Botswana	Radford et al. (2020)
		Flashing light	3	67%	Chile, India	Naha et al. (2020) and Ohrens et al. (2019)
	Sound	Artificial sound	1	100%	Mexico	Zarco-González and Monroy-Vilchis (2014)
	Chemical	Bone oil	2	50%	Canada	Martin and O'Brien (2000)
		Lithium chloride	1	0%	Canada	Bourne and Dorrance (1982)
	Electricity-Physical	Electric fence	3	100%	Canada, Italy, USA	Di Vittorio et al. (2016), Dorrance and Bourne (1980) and Nass and Theade (1988)
	Multimodality-Physical	Guardian dog	5	80%	South Africa	Rust et al. (2013)
		Guardian llamas	3	67%	Australia, USA	Mahoney and Charry (2005) and Meadows and Knowlton (2000)
Elephant	Chemical	Chilli smoke briquette	1	100%	Zambia	Karidozo and Osborn (2015)
		Alternative crop	4	100%	Nepal, Zambia	Gross et al. (2016, 2017)
	Multimodality	Thunder flash	2	100%	Kenya	Sitati and Walpole (2006)
		Chilli pepper fence	1	100%	Zambia	Karidozo and Osborn (2015)
	Chemical-Physical	Combination 1	2	0%	Zimbabwe	Osborn and Parker (2002)
		Combination 2	1	0%	Kenya	Graham and Ochieng (2008)
	Multimodality-Physical	Chilli smoke briquette	1	100%	Zambia	Karidozo and Osborn (2015)
		Alternative crop	4	100%	Nepal, Zambia	Gross et al. (2016, 2017)
	Chemical-Physical	Thunder flash	2	100%	Kenya	Sitati and Walpole (2006)
		Chilli pepper fence	1	100%	Zambia	Karidozo and Osborn (2015)

Note: There was no counterproductive case for these interventions. The ' % stat. – ' represents the proportion of experiments reporting statistically effective evidence to reduce the losses.

TABLE 3 Experimental evidence of interventions for fishery losses caused by marine fauna.

Taxa	Stimuli	Intervention	No. experiments	% stat. –	% stat. +	Country	FAO major area	Reference
Seabird	Physical	Setting funnel	20	10%	5%	Norway	Atlantic, Northeast	Løkkeborg (1998) and Løkkeborg (2001)
		Line weighting	2	50%	0%	Norway	Atlantic, Northeast	Løkkeborg (2001)
		Line weighting	11	9%	0%	Australia, New Zealand, Spain, South Africa, USA, UK	Atlantic, Southeast; Atlantic, Antarctic; Southern Indian Ocean; Mediterranean and Black Sea; Pacific, Northeast; Pacific, Western Central	Agnew et al. (2000), Melvin et al. (2001), Robertson et al. (2006), Dietrich et al. (2008), Melvin et al. (2013), Robertson et al. (2013), Melvin et al. (2014) and Cortes and Gonzalez-Solis (2018)
		Hook-shielding device	1	0%	0%	Australia, Brazil, South Africa	Atlantic, Southwest; Atlantic, Southeast; Indian Ocean, Eastern	Sullivan et al. (2018)
		Line shooter	2	0%	0%	Norway, USA	Atlantic, Northeast; Pacific, Northeast	Melvin et al. (2001) and Løkkeborg and Robertson (2002)
		Mustad lining tube	1	0%	0%	USA	Pacific, Northeast	Melvin et al. (2001)
		Shorter leaded leader	2	0%	0%	Brazil, Uruguay	Atlantic, Southwest	Gianuca et al. (2013) and Jiménez et al. (2013)
		Umbrella-and-stones system	1	0%	100%	Argentina	Atlantic, Southwest	Goetz et al. (2011)
	Light	Blue-dyed bait	2	0%	0%	New Zealand, South Africa	Atlantic, Southeast; Pacific, Southwest	Minami and Kiyota (2004) and Lydon and Starr (2005)
	Multimodality	Artificial bait	1	0%	100%	Spain	Mediterranean and Black Sea	Cortes and Gonzalez-Solis (2018)
Light-Physical		Combination 7	12	42%	0%	USA	Pacific, Northeast	Melvin et al. (2001)
		Toriline	10	40%	0%	Norway, USA	Atlantic, Northeast; Pacific, Northeast	Løkkeborg (1998), Løkkeborg (2001), Melvin et al. (2001), Løkkeborg and Robertson (2002) and Dietrich et al. (2008)
		Combination 6	1	0%	0%	USA	Pacific, Northeast	Melvin et al. (2001))

TABLE 3 (Continued)

Taxa	Stimuli	Intervention	No. experiments	% stat. –	% stat. +	Country	FAO major area	Reference
Elasmobranch	Physical		7	0%	14%			
		Cage sleeve	1	0%	0%	Australia, Fiji	Pacific, Western Central	Hamer et al. (2015)
		Chain device	1	0%	0%	Australia, Fiji	Pacific, Western Central	Hamer et al. (2015)
	Light	Exclusion device	4	0%	25%	Australia	Pacific, Western Central	Robins-Troeger et al. (1995) and Brewer et al. (1998)
		Line weighting	1	0%	0%	USA	Pacific, Eastern Central	Beverly et al. (2009)
		Illuminator	2	0%	0%	Italy, Mexico	Mediterranean and Black Sea; Pacific, Eastern Central	Virgili et al. (2018) and Senko et al. (2022)
		Rare earth metal	1	0%	100%	Canada	Atlantic, Northwest	Godin et al. (2013)
	Electricity	Permanent magnet	2	50%	0%	Australia, USA	Atlantic, Western Central; Pacific, Southwest	O'Connell et al. (2011) and Richards et al. (2018))
	Magnet							
	Multimodality							
Cetacean	Physical		9	33%	0%			
		Squid bait	7	43%	0%	Brazil, USA	Atlantic, Western Central; Atlantic, Eastern Central; Atlantic, Southwest; Atlantic, Southeast	Broadhurst and Hazin (2001), Coelho et al. (2012), Amorim et al. (2015), Driggers III et al. (2016) and Driggers III and Hannan (2019)
		Artificial bait	2	0%	0%		Pacific, Eastern Central	Januma et al. (1999) and Januma et al. (2003)
		Outrigger clip	4	0%	50%	USA	Atlantic, Western Central	Zollett and Read (2006)
		Subsurface netting	2	0%	50%	Australia, Pakistan	Indian Ocean, Western; Pacific, Western Central	Hembree and Harwood (1987) and Kiszka et al. (2021)
	Sound	Umbrella-and-stones system	1	0%	100%	Argentina	Atlantic, Southwest	Goetz et al. (2011)
		Interactive pinger	26	15%	4%	Italy	Mediterranean and Black Sea	Buscaino et al. (2021)
		Pinger	22	50%	0%	Denmark, Germany, Italy, Portugal, Peru, Spain, Turkey, USA	Atlantic, Northeast; Mediterranean and Black Sea; Pacific, Northeast; Pacific, Eastern Central; Pacific, Southeast	Gearin et al. (2000), Culik et al. (2001), Barlow and Cameron (2003), Brotons et al. (2008), Gazo et al. (2008), Buscaino et al. (2009), Larsen et al. (2013), Mangel et al. (2013), Cruz et al. (2014), Larsen and Eigaard (2014), Bilgin and Köse (2018), Snape et al. (2018) and Bruno et al. (2021)
				14%	0%			
				0%	0%			
		Beaded gangion	1	0%	0%			O'Connell et al. (2015)
		Beam chain on net	1	0%	100%	Australia	Pacific, Western Central	Hembree and Harwood (1987)

TABLE 3 (Continued)

Taxa	Stimuli	Intervention	No. experiments	% stat. –	% stat. +	Country	FAO major area	Reference
Pinniped	Physical	Exclusion device	21	14%	29%	Finland, Sweden	Atlantic, Northeast	Lehtonen and Suuronen (2004), Suuronen et al. (2006), Königson et al. (2015) and Calamnius et al. (2018)
			13	23%	31%			Königson et al. (2007)
	Electricity	Fyke net modification	7	0%	29%	Sweden	Atlantic, Northeast	Lunneryd et al. (2003)
		Larger mesh	1	0%	0%	Sweden	Atlantic, Northeast	Forrest et al. (2009)
		Electric wire	1	100%	0%	Canada	Pacific, Northeast	Mate et al. (1986), Barlow and Cameron (2003) Fjälling et al. (2006) and Harris et al. (2014)
		Pinger	4	75%	0%	Sweden, USA, UK	Atlantic, Northeast; Pacific, Northeast; Pacific, Eastern Central	
	Sound							

Note: The '% stat. –' represents the proportion of experiments reporting statistically effective evidence to reduce the losses while the '% stat. +' represents the proportion of experiments reporting statistically counterproductive evidence.

& Osborn, 2015), and decoy field (Hagy et al., 2008) and introduced falcons for farmland birds (Kross et al., 2012).

More evidence not only benefits the implementation of existing interventions but also may help to design new ones (Blackwell et al., 2016; Elmer et al., 2021; Garvey et al., 2020). Although a specific type of stimuli might be more effective than the others within or across animal groups (Barrett et al., 2019; Blackwell et al., 2016; Garvey et al., 2020), we did not find obvious differences either across or within animal groups. This result is in part because the specific stimuli varied greatly in a broad stimulus category. For the electric stimuli, two interventions (i.e. electric fences and wire for carnivores) inducing strong electric shock were effective (Di Vittorio et al., 2016; Forrest et al., 2009; Nass & Theade, 1988) while the rare earth metal to deter elasmobranchs was ineffective (Godin et al., 2013). For the sound stimuli, the distress/alarm call was effective in deterring farmland birds (Khan et al., 2013) while acoustic pingers were generally ineffective in reducing cetacean-caused fishery losses (Brotons et al., 2008). For the chemical stimuli to farmland birds, the outcome of 4-aminopyridine inducing convulsion (Knittle et al., 1988) and methiocarb inducing post-ingestion distress (Tobin et al., 1989) were generally acceptable while the methyl anthranilate was ineffective in all three experiments (Avery et al., 1996). Therefore, we recommend designing new interventions based on these specific stimuli with more effective evidence. Meanwhile, measuring the effects and by-effects of interventions on animals may be needed. For example, improper use of methiocarb can result in avian fatalities (Dolbeer et al., 1994) and electric fences to protect livestock from large carnivores killed more than 1000 pangolins in South Africa (Pietersen, 2022).

More experimental effort is encouraged for animal groups with low management capacity. Our study may establish a baseline for the ongoing effort on these highly concerned animal groups. On the other hand, in practice, there are multifaceted factors affecting the research effort needed, including the severity and frequency of the damage, the amount of economic loss, the income level of the victims, the threatened status of animals and local cultural or religious contexts (Dickman, 2010; Nyhus, 2016). Given the limited resources for conflict management, we highly recommend that future studies develop a general framework to quantify the extent of research effort needed across animal groups.

Surprisingly, the number of experimental studies for terrestrial carnivores and elephants is small though extensive and intensive efforts have been made to manage these conflicts (Montgomery et al., 2022; van Eeden, Eklund, et al., 2018). In fact, many interventions for carnivores and elephants (e.g. electric fences) are too costly for researchers to conduct experiments independently and are often implemented by management authorities and non-profit organizations (or top-down paradigm). The lack of collaboration between researchers and practitioners may contribute to the lack of experimental evaluations for these interventions (van Eeden, Eklund, et al., 2018). Besides, the existing case studies for these costly interventions may be low in statistical power because of small sample sizes (Borenstein et al., 2021). We suggest these

stakeholders work together to provide strong evidence for potential scaling-up efforts.

Meanwhile, we suggest researchers evaluate more replicable and cheap interventions (e.g. flashing lights and chilli fences) to encourage more voluntary uptakes by property owners (or bottom-up paradigm). Additionally, more studies about herbivorous wildfowls (e.g. geese, pheasants and cranes) are needed since they are widely involved in the conflicts (Huang et al., 2023) and the evidence for the related interventions is scarce (Figure 1). Although cetaceans were one of the most studied animal groups in the sea, most of these studies were focused on a single type of intervention (i.e. acoustic pingers), which generally cannot reduce fishery losses. In this study, we were unable to include the conflict-prone ungulates, primates, and molluscivorous and piscivorous birds on aquaculture farms largely because there is no comparable synthesis about interventions for them at a global scale, though many case studies existed (Evers et al., 2022; Russell et al., 2008; Widén et al., 2022). Notably, we found no counterproductive evidence for the interventions on land, which may be partly due to the publication bias based on the outcome of interventions (Shi & Lin, 2019).

In the marine environment, most of the interventions had no impact on fishery losses. Currently, marine fishing activities occur over 55% of the marine surface and many marine faunas frequently interact with fishing gear (Kroodsma et al., 2018; Tixier et al., 2020). Fishing activities are estimated to annually kill nearly 100 million sharks, more than half a million marine mammals, and at least 400,000 seabirds (Read et al., 2006; Worm et al., 2013; Žydelis et al., 2013). Although marine fauna could cause various losses to fisheries, the obligations of interventions in the marine realm are often not merely about reducing the losses but also protecting threatened fauna. In some cases, these two goals can be synergized, such as using grid excluders to reduce seal-caused damages to target catches and seal bycatches (Brewer et al., 1998), but a trade-off for stakeholders is often needed. As such, incentive programs (e.g. eco-labels) and social campaigns (Mulazzani et al., 2021) are encouraged to implement these interventions.

Overall, our results indicated that collective effort is needed to evaluate interventions and launch incentive programs to motivate the implementation of interventions. In practice, these nonlethal technical interventions could be aligned with other categories of interventions (e.g. the lethal and operational) in a sequential mitigation hierarchy (avoiding–minimizing–remediating–offsetting strategy) and produce synergistic effects in an adaptive way. In conclusion, our cross-animal group synthesis could assist stakeholders in tackling the negative human–wildlife interactions outlined in Target 4 of the Kunming–Montreal Global Biodiversity Framework and help improve human–wildlife coexistence.

AUTHOR CONTRIBUTIONS

Cheng Huang conceived the idea; Cheng Huang and Yiqing Chen collected data; Cheng Huang and Yi Wang analysed the data; Cheng Huang, Li-Li Li, Ahimsa Campos-Arceiz, Kenneth O. Onditi, Zhongqiu Li, Limin Feng, Yang Liu, Tien Ming Lee drafted the

manuscript; and all authors revised and approved the final draft of the manuscript.

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CONFLICT OF INTEREST STATEMENT

The authors declare no competing interests. Tien Ming Lee is an Associate Editor of *Journal of Applied Ecology*, but took no part in the peer review and decision-making processes for this paper.

DATA AVAILABILITY STATEMENT

Data available via the Dryad Digital Repository <https://doi.org/10.5061/dryad.wwpzgmsvb> (Huang, Cheng, et al., 2024).

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Figure S1. The inclusion process of case studies.

Table S1. Information on the six syntheses for terrestrial carnivores (van Eeden et al., 2018b; Khorozyan, 2022; Lorand et al., 2022), elephants (Montgomery et al., 2022), farmland birds (Huang et al., 2023), and marine fauna (i.e., pinniped, cetaceans, seabirds, and elasmobranchs; Huang et al., 2024).

Table S2. Detailed interventions of the Combination 1–9.

Table S3. Interventions in sensory stimuli categories.

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